



TCS & AMD AI Hackathon



Basic Information

Team name:

Omar Macías Arévalo

**Names of the team members /
individual / role in the team for the
Hackathon** (Research, coding, building
the business case, presentation, etc)

Omar Macías Arévalo

Project Title:

MediScan AI - Skin lesion detection

Short Description:

Diagnosis of skin lesions (cancer) using artificial intelligence. The user can open the web application on their mobile phone, take a photo of the lesion in real time or upload an image from their gallery so that the application provides a result indicating the type of lesion.

Problem & Context

Problem statement

Diagnosing cancer using medical imaging techniques requires highly trained specialists, who are scarce in resource-limited regions. Demand exceeds the supply of pathologists and dermatologists, leading to delays that reduce survival rates.

Target users / stakeholders

General practitioners, hospitals in developing countries, health centers without specialists

Why this problem matters (business or operational relevance)

Reduce diagnostic time from weeks to seconds. Scale access to remote areas.

Mapped hackathon challenge

AI applied to healthcare on AMD Developer Cloud — Computer vision for medical diagnosis

Solution Overview

**Solution architecture /
workflow diagram**

Medical Imaging - Frontend PWA - FastAPI Backend - EfficientNet-B0
PyTorch - AMD MI300X ROCm GPU

**AI approach used (e.g.,
GenAI, RAG, agents,
vision, speech,
multimodal)**

- Computer vision
- Transfer learning with EfficientNet-B0
- Supervised multi-class tuning
- Class balancing with adaptive weights

**Key technologies /
frameworks leveraged**

- PyTorch 2.12
- EfficientNet-B0
- FastAPI
- PWA (Frontend)
- ROCm 6.4
- AMD MI300X

**What was built during
the hackathon**

MediScan AI - Skin lesion detection

?????

Models used

EfficientNet-B0

Dataset(s) used (if fine-tuning)

HAM10000 (Human Against Machine with 10000 training images).
Source: Kaggle / ISIC Archive

Training time (if fine tuning)

AMD MI300X (192GB HBM3)

- Time per epoch: ~48 seconds.
- Total 20 epochs: ~17 minutes

Number of tokens used for a couple of example scenarios

This project uses computer vision, not LLMs — it does not consume language tokens.

- Classification of 1 image (224×224 px) ~150K pixels processed by the CNN.
- Complete inference ~500ms end-to-end.

End-to-end latency

Image upload ~100ms, Preprocessing (resize, normalize) ~50ms, Inference on AMD MI300X GPU ~150ms, API response (JSON) ~50ms
Total end-to-end ~350-500ms

GPU usage including memory (gives a sense of right sizing)

VRAM used (training)~4GB, VRAM used (inference)~1GB,

Impact & Demo Summary

Expected impact or value (efficiency, productivity, scale, experience)

Reduces diagnosis time from weeks to less than 500 ms. Accessible from any smartphone via PWA without the need for in-store installation. Enables general practitioners to classify skin lesions without requiring a specialist, thus reducing waiting lists. Globally scalable with low operating costs in the AMD cloud.

Key differentiators / innovation

Dermatological AI with EfficientNet-B0 and adaptive class balancing. Complete pipeline: data → model → API → camera-enabled mobile PWA.

Demo flow overview (what the jury should notice)

- 1 Open the app on your smartphone at <http://165.245.136.34:8080>
- 2 Select the camera to take a picture of the injury.
- 3 Select your smartphone's image gallery.
- 4 Tap the Analyze Image button.
- 5 Display the results.



Thank you